

A Quality and disease control system for beekeeping

Christos V. Bellos
Lime Technology IKE
Ioannina, Greece
cbellos@lime-technology.gr

Georgios S. Stergios
Lime Technology IKE
Ioannina, Greece
gstergios@lime-technology.gr

Aleksandros Fyrridis
Lime Technology IKE
Ioannina, Greece
afyrridis@lime-technology.gr

Konstantinos A. Stefanou
Lime Technology IKE
Ioannina, Greece
kstefanou@lime-technology.gr

Sotirios Kontogiannis
University of Ioannina
Ioannina, Greece
skontog@uoi.gr

Abstract—The most significant problems of beekeeping are threats that affect the viability of the bees and the safety risks of beehives, caused by environmental conditions, human interventions, animal attacks and diseases. Several solutions tried to offer sensory monitoring and protection mechanisms, using beehive surveillance camera kits or GPS-based tracking systems. Nevertheless, such systems are not accurate and cannot offer severity estimation and intervention recognition as a holistic approach. The proposed BeeQ project is a complete integrated solution for beehive security and quality control using sensors and low energy transponders placed in the beehive cells of the array. Furthermore, BeeQ incorporates data mining logic to evaluate threats and diseases and inform the beekeeper through the BeeQ mobile application. The main challenges that beekeepers face and BeeQ aims to solve are: (i) Beehive theft, (ii) Bee health problems and colony mortality risks due to illnesses and varroa mite infection that leads to colony destruction, (iii) Extreme weather conditions that may lead to colony losses and extinction, (iv) Animal attacks, (v) Colony losses due to unfortunate farming practices and (vi) Production of high-quality products following production standards.

Keywords; *beekeeping, environmental conditions, IoT sensors, LoRaWAN, safety, data analysis.*

I. INTRODUCTION

The Internet of Things (IoT) industry is shifting at a fast pace towards the agricultural sector, offering new capabilities and assisting in overcoming several barriers of existing solutions. Livestock monitoring is a field that can be affected by the rise of IoT. Thus, BeeQ focuses on livestock monitoring, implementing an IoT management and alert system for modern apiculture [1].

The environment both inside and around the beehives, is of major importance to the success of colony establishment and production [2]. Thus, several solutions tried to offer temperature monitoring and protection mechanisms, with

with beehive surveillance camera kit or GPS-based tracking systems, which are not accurate and cannot offer severity estimation and intervention recognition [3] – [4].

In addition, apiary security is the most important concern of the beekeeping industry [5]. Beehive threats can be categorized into three categories: 1. Human theft and intervention, 2. Mammal attacks and 3. Wasp attacks, mites or swarm colony adaptation incidents such as queen death, extreme conditions, hunger or even thirst. The third category is considered as a conditions security threat with no existing incident response system in the market.

Existing security systems are based either in solely weight tracking systems, cameras or in GPS anti-theft tracking devices and do not offer complete incident response safety solution. Digital weight scales with GPRS or IEEE802.15.4 transponders provide the existing market monitoring and alarm system have trend. However, power consumption and maintenance costs of such a system are significant and in terms of functionality are rather ineffective. For instance, weight scale alarm systems do not really show much potential since several factors like wind, external placement of load cells and load imbalances affect the beehive weight without being relative to safety issues.

Beehive surveillance camera kits are also used in apiculture to track movements and capture photos or records videos and send messages to the beekeeper's mobile phone. This solution may respond to condition security threats, but it introduces significant energy consumption and battery depletion due to the wireless LTE transponders used. For static and moving apiaries electric fence instalments around the beehives are also of practice. However, this solution is not preferred for moving apiaries and offers no effective human theft protection.

Other solutions enforce the installation of GPS anti-theft tracking devices. Beekeepers immediately get notified with GPS coordinates over 3G or SMS for precise mapping of the beehive's location. However, these solutions have high cost for

a complete deployment to all beehives, while in case of animal attacks do not offer any protection since animals only cup side or break the beehives.

BeeQ innovation, besides automated monitoring mentioned earlier, is the safety kit that can be easily installed by the apiarist to existing beehives without modifying them. This goes beyond state of the art and includes a robotic system coupled with low-cost, low-power and robust safety sensors and an incident reporting mechanism, which could identify, assess and cluster the episode on human intervention, animal attack or environmental conditions (i.e., wind). Also, it exploits LoRaWAN gateways offering significantly less energy consumption and cost compared to 3G/GPRS or IEE802.15.4 solutions [6].

The BeeQ system aims to exploit the existing 4000 LoRa gateways in Europe and delivers an incident response system focusing on the safety and monitoring of beehive arrays. It is a complete integrated solution for beehive security and quality control using sensors and low energy transponders placed in the beehive cells of the array. The BeeQ consists of two packages: a) an automated beehive, the Premium package, which includes all sensors and camera pre-installed and pre-configured to communicate with the cloud Information System (BeeQ IS) and b) a ready-to-use safety kit, included in the basic package, which includes only safety mechanisms and is easily attached to existing beehives of the apiarists. Low power and long-range application protocols on top of LoRaWAN wireless protocol, collect data and transmit them to the IS either synchronous or asynchronous depending on the incident. Furthermore, BeeQ incorporates data mining logic to evaluate threats and diseases and inform the beekeeper through the BeeQ mobile application.

II. MATERIALS AND METHODS

BeeQ delivers an integrated automated beekeeping control, management, and safety system for the apiary. It includes sensors and low energy transponders placed in the beehive cells of the array. Low power, low-cost and long-range wireless protocols (LoRaWAN class A), collect the data of each array and transmit them to a cloud Information System (IS), where data analysis is performed, and real-time alerts and monitoring capabilities are offered to apiarists through a mobile application.

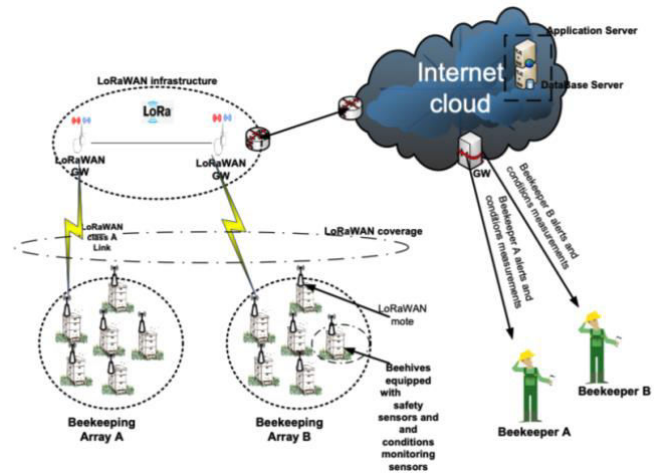


Fig. 1. High level architecture structure of the BeeQ system

The new beehives that are delivered include low price, low energy consumption advanced microelectronics and sensors (AME) for the process of confronting theft and animal hazards as well as environmental conditions (S1-S8). Smart System Integration (SSI) is then applied internally to the beehive (see Fig. 1 - 2), where all AME are being integrated to use a common energy power and transmit simultaneously data to the IS via the existing LoRaWAN BeeQ system infrastructure.

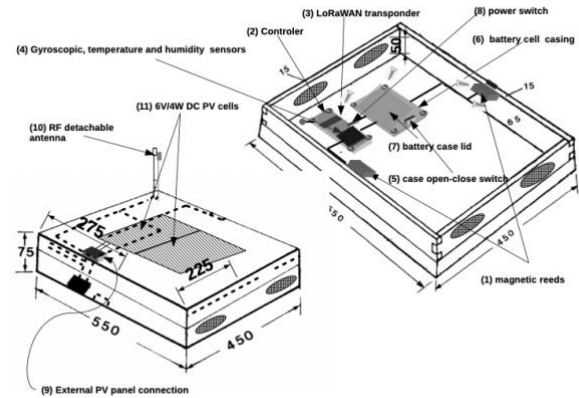


Fig. 2. Automated beehive – lid view

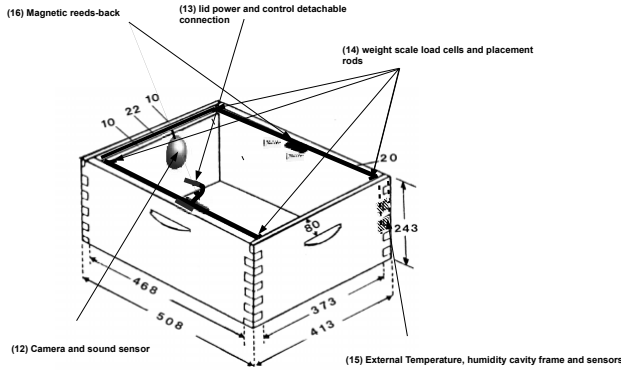


Fig. 3. Automated beehive – box view

S1. Magnetic sensor: A magnetic contact installed on the lid of the automated hives that transmits data to the IS when the lid is opened or misplaced.

S2. Gyroscopic sensor including temperature and humidity sensor internal the hive: Each automated hive has plastic embedded a gyroscopic sensor for detection of shifts, changes in hive position and hive turnovers. Alerts are issued to the cloud IS.

S3. Sound sensor installed in the beehive lid, for the process of monitoring sound level and frequency changes. Bee colony population and living estimations based on sound level-frequency (20Hz - 2KHz) and incidents response, based on changes (queen abandoning the hive, non-mated by a drone queen, food indication signals, piping sound monitoring [7] and distinctive bees stop signals due to wasp and Asian hornets attacks in the beehive).

S4. Load sensors installed inside the hive and under the frames (not externally as current market products enforce with ambiguous results due to load imbalance), for the process of weight monitoring, while current bee weight scale systems install them externally to the hive with the additional risk of being tampered, removed(theft) or malfunction.

S5. internal and external temperature and humidity sensors for the process of monitoring temperature and moisture levels.

S6. Thermo pad surface placed on one side of the brood box for the process of hive temperature regulation and varroa mite destruction.

S8. An IR capable camera with attached led installed inside the hive (with the removal of 1 or 2 frames) for monitoring (still image captures) of worker bees honey depositions, larvae feeding processes and capture images that would lead to disease detection after appliance of pattern recognition algorithms. This camera will not use LoRaWAN to upload images to the cloud IS due to RF duty cycle limitations. The images will be transferred first to the mobile application through BLE transponders incorporated to the LoraWAN capable microprocessor (when the smartphone is close enough

to pair devices) and then uploaded asynchronously to the cloud IS.

On the other hand, the BeeQ basic safety kit includes only S1-S2 (see Fig. 3) plus the LoRaWAN Transponder and battery pack.

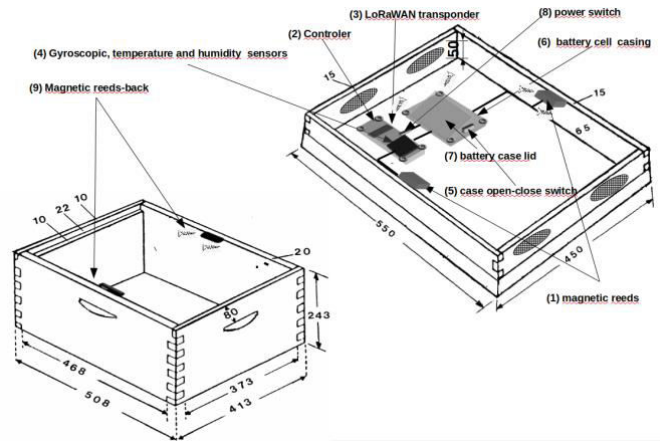


Fig. 4. BeeQ safety kit

Information System (IS) is a web-based portal, acquiring hives' related data (S1-S7) and applying data mining clustering, classification and regression processes to the acquired beehive sensory data streams as well as which type of threats have occurred (i.e. animals, human, environmental) and assess their severity. According to the result, the apiarist will be notified through his BeeQ mobile application. In addition, mask R-CNN image processing, detection and semantic segmentation algorithms will be applied to images captured from the camera (S8) in order to offer (i) periodic Larvae monitoring, (ii) colony queen detection (from the site of the larvae cells), (iii) detection of parasites such as varroa mite on the bees or drones and (iv) Americans epticemia detection (a small bacillus that attacks the bees). Moreover, the IS includes a statistical trends module, to display statistical averages and trends of use, production prediction algorithms and smart agents that will offer cultivation suggestions, guidelines, and suggestive actions.

The Interconnection of the apiarist/farmer with the IS uses a simple and user-friendly functionality, through a mobile application. The apiarist, after authentication, could monitor real-time conditions-related information mentioned earlier. The IS provides specific capabilities to the apiarist through his mobile phone, such as: a) automated information services monitoring the status of the beehives and beehive array events; b) monitoring of the beehives using sensors and supervision services with alerts, notifications, suggestions and forecasts; c) recording of the beekeeper's apicultural interventions within the beehive array (regarding nutrition cases or disease events).

III. DISCUSSION

The EU on beekeeping sector includes 600,000 beekeepers and 16 million hives that is the main target group of BeeQ. Spain, Greece and Italy manage together the 60% of all bee colonies in EU that are managed by professional beekeepers,

which is the second world producer after China, but EU beekeepers have relatively high production costs compared with world competitors.

Potential customers will be targeted through (i) exhibitions of agricultural products, (ii) advertisements and articles in apiculture specific magazines and web portals and (iii) promotional campaigns in social media, specifically Facebook. Lime Technology has already discussed with the Beekeeping Associations of Ioannina and Grevena (acquired Letters of Support), which will support BeeQ and will be part of the testing cases (see Section 1). During the first year of market launch, BeeQ aims to have approximately 50 new customers and attend 2 agricultural exhibitions in a European level. BeeQ targets initially the markets in Spain, Greece and Italy that manage together the 60% of all bee colonies in EU and are leader countries in the number of professional beekeepers. Then, the German market will be targeted, which is the leading EU country in the number of non-professional beekeepers with an estimated number of 8.7 Million bee hives. The cost of the above distribution channels is estimated at 10% of the potential profit.

Afterwards, BeeQ will expand to the rest of Europe, specifically Poland, France, UK, Czech, Romania and Hungary that produce vast amounts of honey aiming to acquire a 10% share of the total European market and be the global leader in the field with its innovative product. A priority will be given to areas that have an existing LoRa Infrastructure, since approximately 4000 LoRa gateways are already installed across Europe. Finally, after Europe, BeeQ will try to expand to Turkey and to China (including Europe are the top three honey producers globally). The main limitation that BeeQ may confront by entering the market is its dependence to LoRaWAN gateways. LoRaWAN has many benefits and this is why it has been chosen by BeeQ as a communication solution, but on the other hand it does not provide a wide-accepted and uniform coverage on the entire European or global area. The existing gateways distribution in northern Europe is a significant advantage, but some premises of apiarists (especially in southern European countries) may not be of dense LoRaWAN coverage. BeeQ will support non-LoRa-covered areas at the second phase of scale-up by covering the cost of installing new gateways to attract new customers (about 500€/gateway covering approximately a circular area of 6-

12km radius). Thus, the profit is estimated to be decreased at 12% in the areas that need LoRaWAN gateways installation. This percentage could be further decreased and the profit of BeeQ could be increased in case two or more customers are being connected to the same LoRaWAN gateway.

BeeQ combines the most enhanced monitoring capabilities with new ideas of smart agents and data mining algorithms on beehive environmental control technologies such as the proposed temperature, humidity, weight and sound monitoring systems, the smart RF-low power sensors security mechanism as well as the incorporation of LoRa technology for long range data transmissions.

For future work the authors aim to emphasize on the evaluation of the developed solution by installing it in different beekeeping premises.

ACKNOWLEDGMENT

This study has been co-financed by the European Union (European Regional Development Fund- ERDF) and Greek national funds through the “BeeQ” Operational Program “Competitiveness, Entrepreneurship & Innovation” (EPAnEK).

REFERENCES

- [1] J. M. Talavera, L. E. Tobon, J. A. Gomez, M. A. Culman, J. M. Aranda, D. T. Parra, L. A. Quiroz, A. Hoyos and L. E. Garreta, “Review of IoT applications in agro-industrial and environmental fields”, *Computer and Electronics in Agriculture*, Vol 42, pp. 283-297, 2017.
- [2] A. T. Mpikos, “What I have to do in my beehives - The real beekeeping”, 2010, ISBN: 978-960- 99316-0-1, in Greek.
- [3] A. Stabentheiner, H. Kovac, and R. Brodschneider, “Honeybee Colony Thermoregulation – Regulatory Mechanisms and Contribution of Individuals in Dependence on Age, Location and Thermal Stress,” *PLoS One*, vol. 5, no. 1, Jan. 2010.
- [4] D. Karakousis, “Apiarist’s experiences”, 2013, Stamoulis publications, ISBN: 9789603519454, in greek.
- [5] H. Clement, “Contemporary Beekeeping”, 2007, Psichalos publications, ISBN: 9789608455344
- [6] The things network, EU LoraWAN coverage map, <https://www.thethingsnetwork.org/map>
- [7] [8] A. Guerrero, T. Anaya, E. Ramirez and C. Tejada, (2017), “Frequency Analysis of Honey Bee Buzz for Automatic Recognition of Health status: A preliminary study”, *Research in Computing Science*, pp. 89-98, ISSN 1870-4096.